

```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c
ograms/0/"semantic
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
int x = 5;
y = 6;
END
declared variable: x
undeclared variable: y
Semantic Analysis Completed.
```

```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c
ograms/0/"semantic
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
int x = 5;
int y = 10;
x = y + 2;
z = x + y;
END
declared variable: x
declared variable: y
undeclared variable: z
Semantic Analysis Completed.
```

```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c
ograms/0/"semantic
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
int x = 5;
if (x > 3) {
    int y = 10;
    z = x + y;
}
y = 20;
END
declared variable: x
Entering new scope (conditional/loop)
undeclared variable: z
Exiting block scope
undeclared variable: y
Semantic Analysis Completed.
ayushi@Ayushis-MacBook-Air 0 %
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```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c
ograms/0/"semantic
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
int m = 10;
m = m
+5;
m = m + 5;
END
declared variable: m
Unknown command: +5;
Semantic Analysis Completed.
```

```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c
ograms/0/"semantic
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
int x = 5;
int x = 10;
END
declared variable: x
Error: Variable 'x' is already declared in this scope
Semantic Analysis Completed.
```

```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c
ograms/0/"semantic
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
float a = 3.14;
int b = 7;
for (int i = 0; i < 10; i++) {
    b = b + i;
}
c = a + b;
END
declared variable: a
declared variable: b
Exiting block scope
undeclared variable: c
Semantic Analysis Completed.
```

```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c -
ograms/0/"semantic
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
int a = 10;
if (a == 10) {
    int b = 20;
    a = a + b;
} else {
    int c = 30;
}
b = a + 5;
END
declared variable: a
Unknown command: } else {
Exiting block scope
undeclared variable: b
Semantic Analysis Completed.
```

```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
int x = 5;
while (x < 10) {
    x = x + 1;
}
y = x + 5;
END
declared variable: x
Entering new scope (conditional/loop)
Exiting block scope
undeclared variable: y
Semantic Analysis Completed.
```

```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
float height = 1.75;
int age = 25;
char initial = 'J';
height = age + 5; // Valid operation
name = "John"; // Error: name is undeclared
END
declared variable: height
declared variable: age
declared variable: initial
undeclared variable: name
Semantic Analysis Completed.
```

```
ayushi@Ayushis-MacBook-Air 0 % cd "/Users/ayushi/c_programs/0/" && gcc semantic.c
Starting Semantic Analysis...
Enter your program (type 'END' to finish):
int count = 0;
float total = 0.0;
char letter = 'A';
if (count < 10) {
    count++;
}
letter = 'B'; // Valid usage
value = total + 1.0; // Error: value is undeclared
END
declared variable: count
declared variable: total
declared variable: letter
Entering new scope (conditional/loop)
Unknown command:      count++;
Exiting block scope
undeclared variable: value
Semantic Analysis Completed.
```